

# Lagrange Interpolation & The Chinese Remainder Theorem

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## Abstract

In this article, we present an application of Lagrange interpolation to solve the puzzle below. Then, we discuss the similarity between Lagrange interpolation and the Chinese remainder theorem.

$$\begin{aligned} 37\#21 &= 928, \\ 77\#44 &= 3993, \\ 123\#17 &= 14840, \\ 71\#6 &= ? \end{aligned}$$

## 1 Lagrange Interpolation

**Theorem 1.1** (Lagrange Interpolation). *Let  $(a_i, b_i)$  be  $n$  distinct pairs of numbers and  $c_i$  be  $n$  not necessarily distinct numbers. By Lemma 1.2 below, we may translate  $(a_i, b_i)$ 's and assume*

$$a_i b_j - a_j b_i \neq 0 \quad \text{for all } j \neq i. \quad (1.1)$$

Define the Lagrange polynomial

$$f(x, y) := \sum_{1 \leq i \leq n} c_i \prod_{1 \leq j \leq n, j \neq i} \frac{x b_j - a_j y}{a_i b_j - a_j b_i}. \quad (1.2)$$

Then,  $f$  is a degree  $d$  homogeneous polynomial with  $d < n$  and  $f(a_i, b_i) = c_i$  for all  $1 \leq i \leq n$ .

**Lemma 1.2.** *Suppose  $a_{i_0} b_{j_0} - a_{j_0} b_{i_0} = 0$  for some  $j_0 \neq i_0$ . By translating  $a_i$ 's and  $b_i$ 's, let*

$$A_{i,s} := a_i + s \quad \text{and} \quad B_{i,t} := b_i + t.$$

*There exists integers  $1 \leq s \leq n$  and  $1 \leq t \leq n(n-1)/2$  such that  $A_{i,s} B_{j,t} - A_{j,s} B_{i,t} \neq 0$  for all  $j \neq i$ .*

*Proof.* If  $a_i \neq 0$  for all  $i$ , set  $s = 0$ . Otherwise, there are at most  $n-1$  non-zero  $a_i$ 's. Choose  $1 \leq s \leq n$  such that  $s \neq -a_i$  for  $1 \leq i \leq n$ . Then,  $A_i \neq 0$  for all  $i$ . We may assume  $a_i \neq 0$  for all  $i$ .

If  $a_{i_0} = a_{j_0} \neq 0$ , then  $b_{i_0} = b_{j_0}$  and these two pairs are not distinct. We may assume  $a_{i_0} \neq a_{j_0}$ . Then,

$$a_{i_0} B_{j_0,t} - a_{j_0} B_{i_0,t} = (a_{i_0} - a_{j_0})t \neq 0 \quad \text{for } 1 \leq t \leq n(n-1)/2.$$

If  $a_i B_{j,t_0} - a_j B_{i,t_0} = 0$  for some  $(i, j) \neq (i_0, j_0)$ , then  $a_i B_{j,t_1} - a_j B_{i,t_1} = (a_i - a_j)(t_1 - t_0) \neq 0$  for any  $t_1 \neq t_0$ . Since there are exactly  $n(n-1)/2 - 1$  pairs of  $i, j$  other than  $i_0, j_0$ , the lemma follows.  $\square$

### 1.1 Solving The Puzzle

The puzzle mentioned in the abstract has  $n = 3$ . Using the notation  $a_i \# b_i = c_i$ , let  $a_1 = 37, b_1 = 21, c_1 = 928, a_2 = 77, b_2 = 44, c_2 = 3993, a_3 = 123, b_3 = 17, c_3 = 14840$ . Then,

$$\begin{aligned} a_1 b_2 - a_2 b_1 &= 37 \cdot 44 - 77 \cdot 21 &= 11, \\ a_1 b_3 - a_3 b_1 &= 37 \cdot 17 - 123 \cdot 21 &= -1954, \\ a_2 b_3 - a_3 b_2 &= 77 \cdot 17 - 123 \cdot 44 &= -4103; \end{aligned}$$

$$\begin{aligned} x \# y &= \frac{c_1(x b_2 - a_2 y)(x b_3 - a_3 y)}{(a_1 b_2 - a_2 b_1)(a_1 b_3 - a_3 b_1)} + \frac{c_2(x b_1 - a_1 y)(x b_3 - a_3 y)}{(a_2 b_1 - a_1 b_2)(a_2 b_3 - a_3 b_2)} + \frac{c_3(x b_1 - a_1 y)(x b_2 - a_2 y)}{(a_3 b_1 - a_1 b_3)(a_3 b_2 - a_2 b_3)} \\ &= 928 \frac{(44x - 77y)(17x - 123y)}{(11)(-1954)} + 3993 \frac{(21x - 37y)(17x - 123y)}{(-11)(-4103)} + 14840 \frac{(21x - 37y)(44x - 77y)}{(1954)(4103)} \\ &= x^2 - y^2. \end{aligned}$$

Finally,  $71\#6 = 71^2 - 6^2 = 5005$ .

## 1.2 The Idea Behind Lagrange Interpolation

The Lagrange polynomial (1.2) looks complicated but the idea behind it actually is simple. Given  $n$  distinct pairs  $(a_i, b_i) \neq (0, 0)$  and  $n$  numbers  $c_i$ , construct a homogeneous polynomial  $f$  satisfying  $f(a_i, b_i) = c_i$ . For discussion purpose, set  $n = 3$ . Let

$$f(x, y) := c_1 \delta_1(x, y) + c_2 \delta_2(x, y) + c_3 \delta_3(x, y),$$

where  $\delta_i$ 's are homogeneous polynomials with the following property:

$$\delta_i(x, y) := \begin{cases} 1 & , \text{ if } x = a_i \text{ and } y = b_i, \\ 0 & , \text{ if } x = a_j \text{ and } y = b_j \text{ for } j \neq i. \end{cases} \quad (1.3)$$

Then,  $f(a_i, b_i) = c_i$  for  $i = 1, 2, 3$  as desired. The remaining task is to construct  $\delta_i$ 's.

Take  $\delta_1$  as an example. Since we want to have  $\delta_1(a_2, b_2) = 0$  and  $\delta_1(a_3, b_3) = 0$ , define it as

$$\delta_1(x, y) := g_2(x, y)g_3(x, y)k_1,$$

where  $g_i(x, y) := xb_i - a_iy$  and  $k_1$  is a constant. Note that  $g_i(a_i, b_i) = 0$  for all  $i$ . Then,

$$\delta_1(x, y) = \begin{cases} g_2(a_1, b_1)g_3(a_1, b_1)k_1 & , \text{ if } x = a_1 \text{ and } y = b_1, \\ 0 & , \text{ if } x = a_j \text{ and } y = b_j \text{ for } j \neq 1. \end{cases}$$

By assumption (1.1), we have  $g_j(a_i, b_i) \neq 0$  for all  $j \neq i$ . In order to have  $\delta_1(a_1, b_1) = 1$ , set  $k_1 := k_{1,2}k_{1,3}$ , where  $k_{i,j} := 1/g_j(a_i, b_i)$  for all  $j \neq i$ . Therefore,

$$\delta_1(x, y) = g_2(x, y)g_3(x, y)k_{1,2}k_{1,3} = \frac{(xb_2 - a_2y)(xb_3 - a_3y)}{(a_1b_2 - a_2b_1)(a_1b_3 - a_3b_1)}$$

satisfies property (1.3). Of course,  $\delta_2$  and  $\delta_3$  can be constructed similarly.

## 2 Chinese Remainder Theorem

**Theorem 2.1** (Chinese Remainder Theorem). *Let  $a_1, \dots, a_n > 1$  be  $n$  pairwise coprime integers. Let  $c_1, \dots, c_n$  and  $m$  be integers such that*

$$m \equiv c_1 \pmod{a_1}, \quad \dots, \quad m \equiv c_n \pmod{a_n}. \quad (2.1)$$

Then,

$$m \equiv \sum_{1 \leq i \leq n} c_i \prod_{1 \leq j \leq n, j \neq i} a_j k_{i,j} \pmod{A}, \quad (2.2)$$

where  $A := \prod_{1 \leq i \leq n} a_i$  and  $k_{i,j}$ 's are integers such that  $k_{i,j} \equiv a_j^{-1} \pmod{a_i}$  for  $j \neq i$ .

### 2.1 The Idea Behind The Chinese Remainder Theorem

Interestingly, the Chinese remainder theorem equation (2.2) looks similar to the Lagrange polynomial (1.2). Indeed, the ideas behind them are essentially the same. As before, we set  $n = 3$  in the following discussion. Let

$$m := c_1 \delta_1 + c_2 \delta_2 + c_3 \delta_3,$$

where  $\delta_i$ 's are integers such that

$$\delta_i \bmod a_j \equiv \begin{cases} 1 & , \text{ if } j = i, \\ 0 & , \text{ if } j \neq i. \end{cases} \quad (2.3)$$

Then,  $m$  satisfies conditions (2.1) as desired. The remaining task is to construct  $\delta_i$ 's.

Take  $\delta_1$  as an example. Since we want to have  $\delta_1 \equiv 0 \pmod{a_2}$  and  $\delta_1 \equiv 0 \pmod{a_3}$ , define it as

$$\delta_1 := a_2 a_3 k_1,$$

for some integer  $k_1$ . Then,

$$\delta_1 \bmod a_j \equiv \begin{cases} a_2 a_3 k_1 & , \text{ if } j = 1, \\ 0 & , \text{ if } j \neq 1. \end{cases}$$

By the assumption of  $a_i$ 's being pairwise coprime, the multiplicative inverses of  $a_2, a_3 \pmod{a_1}$  exist and they are respectively  $k_{1,2}, k_{1,3}$ . In order to have  $\delta_1 \equiv 1 \pmod{a_1}$ , set  $k_1 := k_{1,2}k_{1,3}$ . Therefore,

$$\delta_1 = a_2 a_3 k_{1,2} k_{1,3}$$

satisfies property (2.3). Of course,  $\delta_2$  and  $\delta_3$  can be constructed similarly.